

MSP | MSP-04

# Target Operating Model

Blueprint of the future organisation

## Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

| Field               | Detail                                                             |
|---------------------|--------------------------------------------------------------------|
| Document title      | Target Operating Model                                             |
| Document reference  | MSP-04-HEL-T2-v1.0                                                 |
| Project / Programme | Programme Helios                                                   |
| Version             | 1.0                                                                |
| Status              | Baselined                                                          |
| Date                | 18 Sep 2026                                                        |
| Author              | D. Okonkwo, Programme Manager                                      |
| Owner               | Senior Responsible Owner (developed with Business Change Managers) |
| Approver            | Programme Board                                                    |
| Classification      | Internal                                                           |
| Retention           | 7 years from programme closure                                     |

## Document control

### Version history

| Version | Date        | Author                        | Summary of change                        |
|---------|-------------|-------------------------------|------------------------------------------|
| 0.1     | 04 Jun 2026 | D. Okonkwo, Programme Manager | Initial draft                            |
| 0.2     | 21 Jul 2026 | D. Okonkwo, Programme Manager | Updated following workstream lead review |
| 1.0     | 18 Sep 2026 | D. Okonkwo, Programme Manager | Baselined at the Tranche 2 funding gate  |

### Reviewers

| Name                                   | Role                     | Review scope                              | Date reviewed |
|----------------------------------------|--------------------------|-------------------------------------------|---------------|
| T. Nakamura, Chief Platform Architect  | Chief Platform Architect | Technical content and solution alignment  | 08 Sep 2026   |
| L. Haddad, CISO                        | CISO                     | Security, assurance and residency content | 10 Sep 2026   |
| C. Whitfield, Finance Business Partner | Finance Business Partner | Cost, funding and benefit figures         | 11 Sep 2026   |
| M. Verity, Head of P3O                 | Head of P3O              | Governance standards and completeness     | 15 Sep 2026   |

### Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

| Name                                   | Role                                 | Decision                               | Date        |
|----------------------------------------|--------------------------------------|----------------------------------------|-------------|
| R. Al Mazrouei, Group CTO              | Senior Responsible Owner / Group CTO | Approved                               | 18 Sep 2026 |
| H. Farrell, Director of Infrastructure | Director of Infrastructure           | Approved                               | 18 Sep 2026 |
| L. Haddad, CISO                        | CISO                                 | Approved subject to closure of ISS-003 | 18 Sep 2026 |

### Related products

| Reference | Product          | Relationship                  |
|-----------|------------------|-------------------------------|
| MSP-03    | Vision Statement | Source                        |
| MSP-05    | Benefits Map     | Traceability                  |
| MSP-10    | Projects Dossier | Derived from the gap analysis |

## How to use this template

| Aspect             | Guidance                                                                                                                                                                           |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Describes in detail the future state the programme will deliver, covering processes, organisation, technology, information and culture, so that projects can be scoped against it. |
| Framework          | MSP — Design theme; created in Define the Programme; refined at each tranche boundary                                                                                              |
| Created when       | Define the Programme                                                                                                                                                               |
| Reviewed / updated | At each tranche boundary                                                                                                                                                           |
| Owner              | Senior Responsible Owner (developed with Business Change Managers)                                                                                                                 |
| Approver           | Programme Board                                                                                                                                                                    |
| Format options     | Document, presentation, register entry, or tool record — the content matters, not the medium.                                                                                      |
| Tailoring          | Scale the depth of each section to the scale, risk and complexity of the work. Delete sections that add no value and record the tailoring decision in the governance approach.     |

*This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.*

### Quality criteria

*Use these to check the product is fit for purpose before submitting it for approval.*

- Every element of the future state is traceable to a benefit
- The current state is described with the same rigour as the future state
- Gaps between current and future state can be turned into project scope
- Interim states at each tranche boundary are described and are individually operable

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## 1. Purpose and scope of the model

*In scope: the physical infrastructure in DXB-1 halls A and B, the compute and storage estate, the network fabric, the orchestration and scheduling platform, the MLOps and developer experience layer, the security controls required for multi-tenancy, and the operational readiness required to run the estate. Out of scope: the models themselves, business-unit data engineering, the AUH-2 second site (Tranche 3), and any change to the existing corporate WAN.*

## 2. Current state

### Processes

| Process                               | Current description                                                                           | Performance today                                                | Pain points                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Requesting AI compute capacity        | Email or ticket to a departmental infrastructure team; manual approval by the department head | 23 working days median from request to usable environment        | Opaque queue, no visibility of position, inconsistent standards between departments      |
| Scheduling and running training jobs  | Each department runs its own scheduler; jobs cannot span departments                          | 31% average utilisation; jobs above 32 GPUs cannot be run at all | Capacity stranded in silos; large jobs impossible; no pre-emption                        |
| Onboarding a new workload             | Bespoke environment built by hand for each workload                                           | 4 to 6 weeks per workload                                        | No reproducibility; environments drift; security review repeated each time               |
| Recovering compute cost               | Cost absorbed centrally in the technology budget; no visibility to consumers                  | No consumption data available                                    | No incentive to use capacity efficiently; no basis for business unit budgeting           |
| Responding to an infrastructure fault | Departmental teams respond in hours during business hours only                                | No formal restoration target                                     | No coverage for liquid cooling; no 24x7 capability; hardware at risk from coolant events |

### Organisation

| Function / team                   | Structure today                                                    | Headcount | Capability gaps                                               |
|-----------------------------------|--------------------------------------------------------------------|-----------|---------------------------------------------------------------|
| Departmental infrastructure teams | Four separate teams of 3 to 5, each running a departmental cluster | 16        | No liquid cooling, GPU scheduling or multi-tenancy experience |
| Platform engineering              | Does not exist as a distinct function                              | 0         | Entire capability absent                                      |
| Service operations                | Air-cooled data centre operations, 5x8 coverage                    | 12        | No liquid cooling competence; no 24x7 for this estate         |
| Group Security                    | Reviews architecture at gate points                                | 9         | Limited cloud-native and supply chain security depth          |
| AI Research and Fraud Analytics   | Manage their own infrastructure alongside research work            | 31        | Time lost to infrastructure work rather than research         |

## Technology and systems

| System / platform             | Purpose                                                  | Condition                                                  | Constraints                                                       |
|-------------------------------|----------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------|
| Departmental GPU clusters (4) | Training and inference for a single department each      | Two are out of vendor support; all are at capacity         | Cannot be pooled; no multi-tenancy; inconsistent security posture |
| Public cloud GPU rental       | Burst capacity and workloads the departments cannot host | In use, growing at ~40% per year                           | Residency exceptions required; unit cost is high                  |
| Helios compute platform       | Shared multi-tenant AI training and inference            | Being built                                                | DXB-1 capped at 6.2 MVA until 2028                                |
| Parallel filesystem tier      | High-throughput storage for training datasets            | Being built                                                | Metadata tier sizing drives small-file performance                |
| Enterprise identity platform  | Authentication and authorisation across the Group        | In service; workload identity federation not yet delivered | Federation epic deprioritised by the owning programme (DEP-003)   |

## Information and data

| Information asset                          | Owner                                 | Current handling                                        | Issues                                           |
|--------------------------------------------|---------------------------------------|---------------------------------------------------------|--------------------------------------------------|
| Training datasets — fraud analytics corpus | M. Duarte, Head of Fraud Analytics    | Held on departmental storage; classification incomplete | Cannot be onboarded until classified (ISS-007)   |
| Model weights and checkpoints              | Dr S. Iyer, Head of AI Research       | Stored on local scratch with no retention policy        | No lineage; checkpoints lost between experiments |
| Consumption and utilisation telemetry      | G. Lindqvist, Service Operations Lead | Not collected                                           | No basis for benefit measurement or chargeback   |
| Container images and SBOMs                 | K. Adeyemi, Security Architect        | Images pulled directly from public registries           | No provenance; no admission control              |
| Assurance and control evidence             | L. Haddad, CISO                       | Produced manually for each review                       | Expensive and point-in-time only                 |

## Culture, behaviours and ways of working

### 3. Future state

#### Processes

| Process                        | Future description                                                                   | Target performance          | Enabling change                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------|
| Requesting AI compute capacity | Self-service request through the platform portal against a pre-approved tenant quota | Under 2 hours, self-service | Quota model agreed, portal built, approval delegated to tenant quota rather than |

| Process                               | Future description                                                               | Target performance                                                           | Enabling change                                                                      |
|---------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|                                       |                                                                                  |                                                                              | per-request                                                                          |
| Scheduling and running training jobs  | Single scheduling plane with fair-share quota, pre-emption and checkpoint-resume | 72% utilisation; jobs to 256 GPUs supported                                  | Single scheduler decision (DEC-003); departmental clusters decommissioned            |
| Onboarding a new workload             | Templated environment from a signed base image, with security controls inherited | Under 3 days per workload                                                    | Image supply chain built; security controls moved into the platform baseline         |
| Recovering compute cost               | Per-tenant metering with a published rate card and monthly statements            | Statements reconciling to telemetry within 2%                                | Rate card approved (DEC-006); metering delivered; twelve-month subsidised transition |
| Responding to an infrastructure fault | NOC-monitored with automated leak isolation and a 24x7 on-call rota              | NOC alerted within 60 seconds; four-hour restoration for scheduled workloads | Runbooks written, two live drills, on-call rota established, vendor cover contracted |

## Organisation

| Function / team                   | Future structure                                                                          | Headcount | New capabilities                                                        |
|-----------------------------------|-------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------|
| Departmental infrastructure teams | Consolidated into a single platform run team; specialists redeployed to tenant enablement | 16        | Platform SRE, GPU scheduling, capacity management                       |
| Platform engineering              | Nine-person platform run team owning scheduling, quota, images and observability          | 0         | Kubernetes at scale, policy-as-code, platform product management        |
| Service operations                | Extended to 24x7 for the Helios estate with liquid-cooling trained staff                  | 12        | Coolant leak response, high-density power, GPU telemetry interpretation |
| Group Security                    | Security architect embedded in the platform team as a design participant                  | 9         | Admission control policy, attestation, tenant isolation design          |
| AI Research and Fraud Analytics   | Consume the platform as tenants; no infrastructure responsibility                         | 31        | Platform-native workflows, experiment tracking discipline               |

## Technology and systems

| System / platform             | Future role                                      | Change required                                                                | Owner                                  |
|-------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------|
| Departmental GPU clusters (4) | Decommissioned                                   | Workloads migrated to Helios, then hardware disposed of under the asset policy | H. Farrell, Director of Infrastructure |
| Public cloud GPU rental       | Retained for burst only, at approximately 25% of | Committed workloads migrate to Helios; burst policy published                  | C. Whitfield, Finance Business Partner |

| System / platform            | Future role                                                  | Change required                                                | Owner                                  |
|------------------------------|--------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------|
|                              | <i>current spend</i>                                         |                                                                |                                        |
| Helios compute platform      | Primary AI compute platform for the Group                    | Tranches 1 to 3                                                | P. Raghavan, Platform Engineering Lead |
| Parallel filesystem tier     | 12 PB usable at 1.2 TB/s, expanding in Tranche 3             | Built and accepted against a workload-representative benchmark | I. Petrova, Data Platform Lead         |
| Enterprise identity platform | Provides workload identity for all platform service accounts | Dependency escalated to the Portfolio Board                    | K. Adeyemi, Security Architect         |

## Information and data

| Information asset                          | Future handling                                                                      | Governance requirement                                 | Owner                                 |
|--------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------|---------------------------------------|
| Training datasets — fraud analytics corpus | Held on the parallel filesystem within the tenant namespace, classified and labelled | Restricted — must remain in UAE at rest and in transit | M. Duarte, Head of Fraud Analytics    |
| Model weights and checkpoints              | Versioned artefacts in the model registry with lineage to dataset version            | Confidential — per-tenant KMS keys                     | Dr S. Iyer, Head of AI Research       |
| Consumption and utilisation telemetry      | 10-second resolution telemetry with 13-month retention                               | Internal — retained 13 months                          | G. Lindqvist, Service Operations Lead |
| Container images and SBOMs                 | Signed images in the internal registry with SBOM attached                            | Internal — provenance evidence retained 7 years        | K. Adeyemi, Security Architect        |
| Assurance and control evidence             | Generated automatically by the pipeline and control plane                            | Confidential — retained 7 years for audit              | L. Haddad, CISO                       |

## Culture, behaviours and ways of working

### 4. Gap analysis

The difference between current and future state, expressed so that each gap can be assigned to a project or business change activity.

| Ref   | Dimension      | Gap                                                      | Change required                                   | Assigned to                                               | Benefit enabled                                             |
|-------|----------------|----------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------|
| GA-01 | Organisational | No immutable dataset versioning in the training pipeline | Add coolant leak detection integration to the NOC | Aligned to the strategic theme of sovereign AI capability | Reduces median training cycle time from 11.5 days to 3 days |
| GA-02 | Regulatory     | No fair-share quota mechanism exists on the              | Reduce Tranche 2 GPU count from 2,048 to 1,792    | Maintained by the P3O as                                  | Removes four standing data                                  |



| Ref   | Dimension  | Gap                                                               | Change required                                                   | Assigned to                                                       | Benefit enabled                                                    |
|-------|------------|-------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------|
|       |            | <i>pilot cluster</i>                                              |                                                                   | <i>part of the assurance evidence library</i>                     | <i>residency exceptions</i>                                        |
| GA-03 | Technical  | <i>Firmware attestation API not yet available from the vendor</i> | <i>Add a second CDU to hall A to reach the 60 kW design point</i> | <i>Reflected in the solution intent as a fixed constraint</i>     | <i>Reduces median training cycle time from 11.5 days to 3 days</i> |
| GA-04 | Facilities | <i>Second-site failover untested</i>                              | <i>Extend contractor SRE cover by six months</i>                  | <i>Reviewed at each PI boundary by the Release Train Engineer</i> | <i>Lifts average GPU utilisation from 31% to 72%</i>               |

## 5. Interim states by tranche

Each interim state must be a coherent, operable configuration of the organisation — not a half-finished one.

| Tranche   | Interim state description                                                                    | Capabilities delivered | Benefits enabled                                                   | Operable?                                                                |
|-----------|----------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------|
| Tranche 1 | <i>Foundation — 256-GPU pilot cluster in DXB-1, proving power, cooling and fabric design</i> | No                     | <i>Reduces median training cycle time from 11.5 days to 3 days</i> | <i>Applies from hall A handover; hall B follows in Tranche 2 closure</i> |
| Tranche 2 | <i>Scale — 2,048 GPUs across DXB-1 halls A and B, production platform and MLOps stack</i>    | Yes                    | <i>Removes four standing data residency exceptions</i>             | <i>Agreed with the integration partner at design freeze, 15 Oct 2026</i> |
| Tranche 3 | <i>Multi-tenancy and resilience — 4,096 GPUs, AUH-2 second site, chargeback and DR</i>       | Partially              | <i>Reduces median training cycle time from 11.5 days to 3 days</i> | <i>Applies only to workloads classified Restricted or above</i>          |

## 6. Design constraints and assumptions

| Ref     | Constraint / assumption                                                                                 | Source                                    | Impact if changed                                                                             |
|---------|---------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------|
| ASM-001 | <i>Vendor GPU allocation of 2,048 units for Tranche 2 will be honoured at the quoted delivery dates</i> | <i>Vendor allocation letter, Jun 2026</i> | <i>Tranche 2 completion slips by at least one quarter and contract terms are renegotiated</i> |

| Ref     | Constraint / assumption                                                                                 | Source                                  | Impact if changed                                                                        |
|---------|---------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------|
| ASM-002 | <i>DXB-1 has 6 MVA of usable capacity available to the programme without a utility upgrade</i>          | <i>Site capacity study, May 2026</i>    | <i>Tranche 2 capped at 1,024 GPUs and the AUH-2 site is brought forward two quarters</i> |
| ASM-003 | <i>Three business units will migrate committed workloads within six months of hall A handover</i>       | <i>Business case, Apr 2026</i>          | <i>Utilisation benefit of AED 14.2m over five years is not realised in full</i>          |
| ASM-004 | <i>Training data classified as Restricted must remain within UAE borders at rest and in transit</i>     | <i>Group data residency policy v4.2</i> | <i>Not applicable — this is a fixed constraint on design</i>                             |
| ASM-005 | <i>Liquid cooling maintenance can be absorbed by the existing facilities contract without an uplift</i> | <i>FM contract review, Jun 2026</i>     | <i>Annual run cost increases by approximately AED 780,000</i>                            |

## 7. Design authority and change control

*Who owns the integrity of the model, and how proposed departures from it are assessed and decided.*

*Programme Helios — Tranche 2 scale build, DXB-1 halls A and B. Aligned to the strategic theme of sovereign AI capability. Detail is maintained by the Senior Responsible Owner (developed with Business Change Managers) and reviewed at the monthly Programme Board.*